

```
import cv2
import numpy as np
from google.colab import files
from google.colab.patches import cv2_imshow

# Upload image
uploaded = files.upload()

# Get uploaded filename
filename = next(iter(uploaded))

# Read image
img = cv2.imread(filename)

# Check if image loaded
if img is None:
    print("Error: Image could not be loaded.")
else:
    # Laplacian kernel
    kernel = np.array([
        [1, 1, 1],
        [1, -8, 1],
        [1, 1, 1]
    ])

    # Apply filter
    sharpened = cv2.filter2D(img, -1, kernel)

    # Display original image
    print("Original Image:")
    cv2_imshow(img)

    # Display filtered image
    print("Filtered / Sharpened Image:")
    cv2_imshow(sharpened)

    # Save output
    cv2.imwrite("Sharpened_Image.jpg", sharpened)

    print("Sharpened_Image.jpg saved successfully.")
```

Choose Files sampled_image.jpg

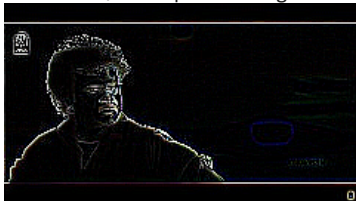
sampled_image.jpg(image/jpeg) - 9699 bytes, last modified: 7/22/2026 - 100% done

Saving sampled_image.jpg to sampled_image (6).jpg

Original Image:



Filtered / Sharpened Image:



Sharpened_Image.jpg saved successfully.